COMPARISON OF THREE CHIMP,GORILLA,ACTF:

FOR DATASET:

```
import numpy as np

plateau1 = [200.0] * 40

plateau2 = list(200.0 + 1e-7*(i%3) for i in range(30))

alt_pattern = [300.0 if i%2==0 else 320.0 for i in range(40)]

sparse0 = [0.0] * 18 + [1.0, 0.0]

plateau3 = [105.0]*10 + [109.0]*5 + [105.0]*15

test_data = plateau1 + plateau2 + alt_pattern + sparse0 + plateau3

print(f"Total points: {len(test_data)}")
```

```
PS C:\Users\suvat\Downloads\Data> py comparison.py
FLOATING-POINT COMPRESSION ALGORITHM COMPARISON
SPECIALIZED TEST DATASET
================================================================================
Dataset composition:
  Plateau 1: 40 identical values (200.0)
  Plateau 2: 30 values with micro-variations (200.0 + 1e-7*(i%3))
  Alt Pattern: 40 alternating values (300.0/320.0)
  Sparse: 18 zeros + [1.0, 0.0]
  Plateau 3: Mixed plateaus [105,109,105]
  Total points: 160

================================================================================
COMPRESSION COMPARISON: Mixed Pattern Dataset
================================================================================

DATA ANALYSIS:
  Total values: 160
  Identical consecutive: 84 (52.8%)
  Small changes (≥32 leading zeros): 29 (18.2%)
  Large changes (<32 leading zeros): 46 (28.9%)

COMPRESSION RESULTS:
Algorithm    Time (ms)  Compressed Bits  Ratio (%)    Bits/Value   Rating
---------------------------------------------------------------------------
Gorilla      0.464      3189             68.86        19.93        GOOD
Chimp        0.591      1807             82.35        11.29        BETTER
ACTF         0.660      1690             83.50        10.56        BEST

TECHNICAL SUMMARY:
  Original size: 10240 bits (1280.0 bytes)
  Best actual compression: ACTF (83.50% ratio)
  Fastest actual: GORILLA (0.464 ms)
```

```
ALGORITHM ASSESSMENT:
  🟢 ACTF: ADVANCED COMPRESSION - Most sophisticated algorithm with multiple
          compression modes. Best for complex real-world datasets.
  🟡 Chimp: GOOD PERFORMANCE - Optimized for time series but limited
          in handling mixed patterns.
  🔴 Gorilla: BASIC COMPRESSION - Simple and reliable but lacks advanced
          optimization techniques.

🎯 FINAL RECOMMENDATION:
  USE ACTF COMPRESSION
  - Most advanced compression techniques
  - Best handling of mixed data patterns
  - Future-proof architecture
  - Superior real-world performance

================================================================================
DETAILED COMPRESSION BREAKDOWN
================================================================================
Original data requires: 10240 bits
Gorilla compressed to:  3189 bits
Chimp compressed to:    1807 bits
ACTF compressed to:     1690 bits

KEY ADVANTAGES OF ACTF:
  ✓ Multiple compression modes for different data patterns
  ✓ Better handling of alternating value sequences
  ✓ Superior compression of mixed plateau data
  ✓ Advanced state management for complex patterns
  ✓ Optimal for real-world time series with varying characteristics
```

HERE YOU CAN SEE THAT ACTF PERFORMED EXCELLENTLY THAN OTHER TWO..